

Rank-one Toeplitz semicommutators vanish on the unit ball

The several-variable half of the possible-rank problem

Autonomous solve-first investigation, lane 12

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Abstract

Le and Tikaradze ask whether the Ding–Qin–Zheng possible-rank theorem for a single product of Toeplitz operators on the disc extends to several variables. We prove the rank-one half. On every weighted Bergman space $A_\gamma^2(\mathbb{B}_N)$ with $N \geq 2$ and nonnegative integer γ , under the source theorem’s boundedness and smoothness assumptions, a difference $T_\phi T_\psi - T_h$ of rank at most one must be zero. The proof converts the source’s Brown–Halmos identity into a doubly centered holomorphic kernel. Its alleged separation rank is at most one, whereas a coefficient-tensor estimate gives the lower bound

$$\binom{2N+1+\gamma}{N} - 2 \geq 8.$$

The construction and classification of nonzero ranks at least two remain open in this packet.

1 Question and result

Let $\mathbb{B}_N$ be the unit ball in $\mathbb{C}^N$. For a bounded symbol a , let T_a denote the Toeplitz operator on $A_\gamma^2(\mathbb{B}_N)$. Throughout, $\gamma \in \{0, 1, 2, \dots\}$. The main theorem of Le–Tikaradze [1] states, in the case of one product, that if $\phi = f + \bar{g}$ and $\psi = u + \bar{v}$ are bounded pluriharmonic functions, h is bounded and $C^{2N+2+2\gamma}$, and

$$T_\phi T_\psi = T_h + \sum_{\ell=1}^r x_\ell \otimes y_\ell, \tag{1}$$

then, writing $s = N + 1 + \gamma$,

$$h - \bar{g}u \text{ is pluriharmonic,} \quad \phi\psi = h + (1 - |z|^2)^s \sum_{\ell=1}^r x_\ell \bar{y}_\ell. \tag{2}$$

The converse also holds. Here $x \otimes y$ maps q to $\langle q, y \rangle x$.

In their Remark 4.3, the authors recall that Ding–Qin–Zheng [2] proved in one variable that rank one in (1) is impossible, while every rank $r \geq 2$ occurs. They ask for a several-variable generalization.

Theorem 1 (rank-one vanishing). *Let $N \geq 2$ and let γ be a nonnegative integer. Suppose ϕ, ψ are bounded pluriharmonic functions on $\mathbb{B}_N$ and h is bounded and $C^{2N+2+2\gamma}$. If*

$$\text{rank}(T_\phi T_\psi - T_h) \leq 1,$$

then $T_\phi T_\psi = T_h$.

Thus the first half of the Ding–Qin–Zheng possible-rank theorem survives in every complex dimension at least two and for every integer weight covered by the source. The proof actually gives an eight-dimensional obstruction in the smallest case $N = 2$, $\gamma = 0$.

2 A centered tensor-rank lemma

For a holomorphic function q on $\mathbb{B}_N$, put $Qq = q - q(0)$. If

$$K(z, w) = \sum_{j=1}^m a_j(z) \overline{b_j(w)},$$

its *separation rank* $\text{srnk } K$ is the least length of such a representation. Equivalently, it is the rank of the associated finite coefficient tensor.

Lemma 2. *Let x, y be nonzero holomorphic functions on $\mathbb{B}_N$, let $s \geq 1$, and put*

$$R(z, w) = Q_z Q_w [(1 - \langle z, w \rangle)^s x(z) \overline{y(w)}].$$

Then

$$\text{srnk } R \geq \binom{N+s}{N} - 2. \quad (3)$$

Proof. Write $M = \binom{N+s}{N}$, the number of multi-indices α with $|\alpha| \leq s$. The multinomial expansion is

$$(1 - \langle z, w \rangle)^s = \sum_{|\alpha| \leq s} c_\alpha z^\alpha \overline{w^\alpha}, \quad c_\alpha = (-1)^{|\alpha|} \binom{s}{|\alpha|} \frac{|\alpha|!}{\alpha!}, \quad (4)$$

and every c_α is nonzero.

Let E be the M -dimensional coefficient space with basis e_α and define

$$Ae_\alpha = z^\alpha x, \quad Be_\alpha = z^\alpha y.$$

Both A and B are injective. For example, $\sum d_\alpha z^\alpha x = 0$ forces the polynomial $\sum d_\alpha z^\alpha$ to vanish on the nonempty open set where $x \neq 0$, and hence identically.

Put $U = QA$ and $V = QB$. Since A is injective and the space of constant functions is one-dimensional,

$$\dim \ker U \leq 1, \quad \dim \ker V \leq 1. \quad (5)$$

Let $C : E \rightarrow E$ be diagonal with diagonal entries c_α . Expansion (4) says that the coefficient tensor of R is $UCV^\top$. The map C is invertible. Applying Sylvester's rank inequality in the finite-dimensional images of U and V gives

$$\text{srnk } R = \text{rank}(UCV^\top) \geq \text{rank } U + \text{rank } V - M \geq (M-1) + (M-1) - M = M-2.$$

This proves (3). $\square$

The two lost dimensions in the lemma are genuine possibilities: centering may remove a constant lying in either monomial-multiple span. No independence between Qx and the positive-degree monomial multiples is silently assumed.

3 Proof of rank-one vanishing

Proof of Theorem 1. Assume for contradiction that

$$T_\phi T_\psi - T_h = x \otimes y \quad (6)$$

for nonzero $x, y \in A_\gamma^2(\mathbb{B}_N)$. In the decompositions $\phi = f + \bar{g}$ and $\psi = u + \bar{v}$, constants may be transferred between the holomorphic and antiholomorphic summands, so we normalize $f(0) = v(0) = 0$.

Apply (2) to (6). Since

$$\begin{aligned} h - \bar{g}u &= \phi\psi - \bar{g}u - (1 - |z|^2)^s x\bar{y} \\ &= fu + f\bar{v} + \bar{g}\bar{v} - (1 - |z|^2)^s x\bar{y} \end{aligned}$$

is pluriharmonic, and the first and third terms on the last line are holomorphic and antiholomorphic respectively, the function

$$f\bar{v} - (1 - |z|^2)^s x\bar{y}$$

is pluriharmonic.

Complexify the real-analytic identity by replacing $\bar{z}$ with an independent $\bar{w}$. The complexification of a pluriharmonic function has no terms of positive holomorphic and antiholomorphic degree; applying $Q_z Q_w$ therefore kills it. The normalization of f and v yields

$$f(z)\overline{v(w)} = Q_z Q_w [(1 - \langle z, w \rangle)^s x(z)\overline{y(w)}]. \quad (7)$$

The left side has separation rank at most one. Lemma 2 says that the right side has separation rank at least

$$\binom{N+s}{N} - 2 = \binom{2N+1+\gamma}{N} - 2.$$

For $N \geq 2$ and $\gamma \geq 0$, the minimum is $\binom{5}{2} - 2 = 8$. This contradicts (7). Hence no nonzero rank-one difference exists. $\square$

The same coefficient argument also rules out rank one when $N = 1$ and $\gamma \geq 1$. In the sole borderline case $N = 1$, $\gamma = 0$, its lower bound is only one; that is precisely the case requiring the sharper one-variable argument of [2].

4 Boundary of the result

The theorem settles the rank-one half of the question, not the entire possible-rank program. For rank $r > 1$, equation (7) is replaced by a sum over x_ℓ, y_ℓ . Although minimal rank representations make each family $\{x_\ell\}$ and $\{y_\ell\}$ linearly independent over $\mathbb{C}$, their monomial multiples can have polynomial-module relations such as $x_2 = z_1 x_1$. Those relations can cancel the diagonal blocks used in Lemma 2.

Eight upgrade routes were tested. In particular, a prospective finite-rank divisibility lemma for $(1 - \langle z, w \rangle)^s K(z, w)$ would finish the classification, but the obvious zero-multiplicity proof is invalid because the hypersurface $\langle z, w \rangle = 1$ lies outside the product of unit balls and the holomorphic factors need not continue there. Lifting the one-variable examples through one coordinate likewise creates mixed monomials in all remaining coordinates. No controlled construction of every rank $r \geq 2$, nor a proof that all nonzero finite ranks are impossible, is claimed.

A bounded search on 13 August 2026 covered the exact quoted question, arXiv id and title, later citations, and rank-one/semicommutator terminology. It found the source and the one-variable theorem, but no later several-variable resolution. Novelty is plausible, not certified.

References

- [1] T. Le and A. Tikaradze, *A generalization of the Brown–Halmos theorems for the unit ball*, Adv. Math. 404 (2022), 108411; arXiv:2101.03937.
- [2] X. Ding, Y. Qin, and D. Zheng, *A theorem of Brown–Halmos type on the Bergman space modulo finite rank operators*, J. Funct. Anal. 273 (2017), 2815–2845.